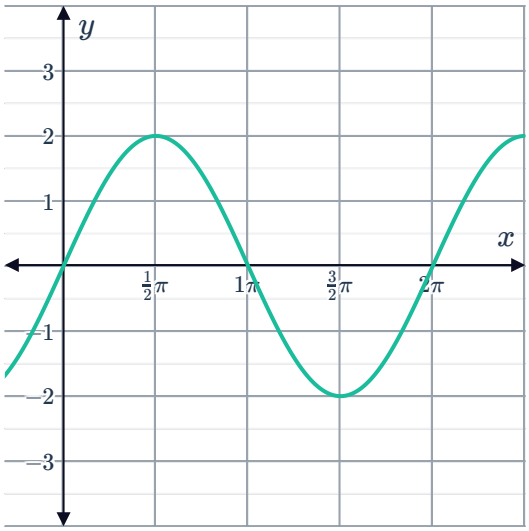
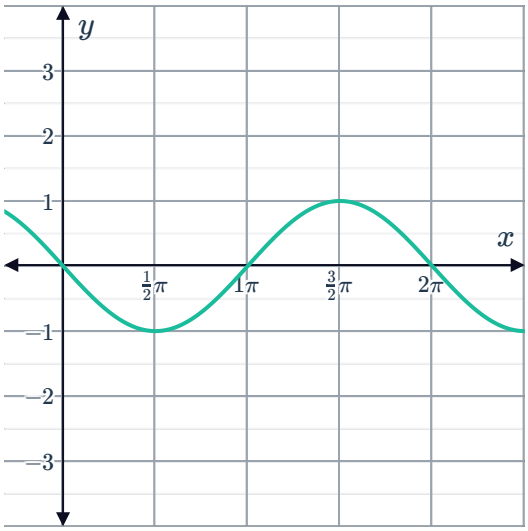
5 State the equation of each of the functions graphed below given that they are of the form $y = a\sin x$ or $y = a\cos x$:

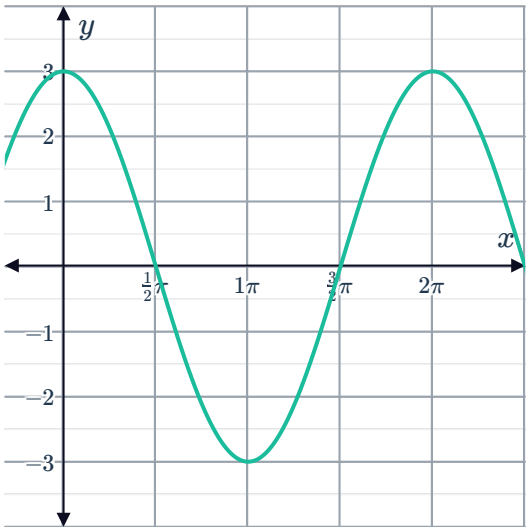
a



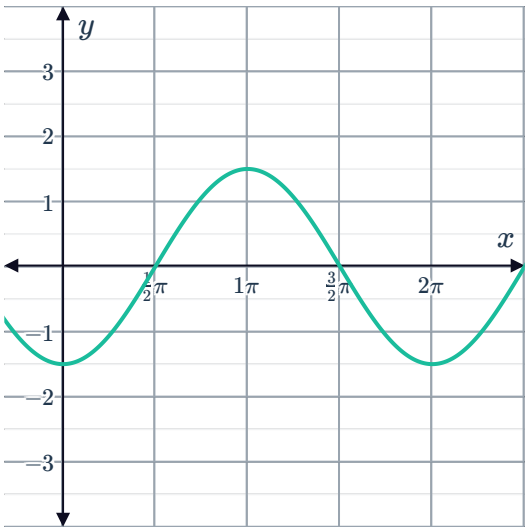
b



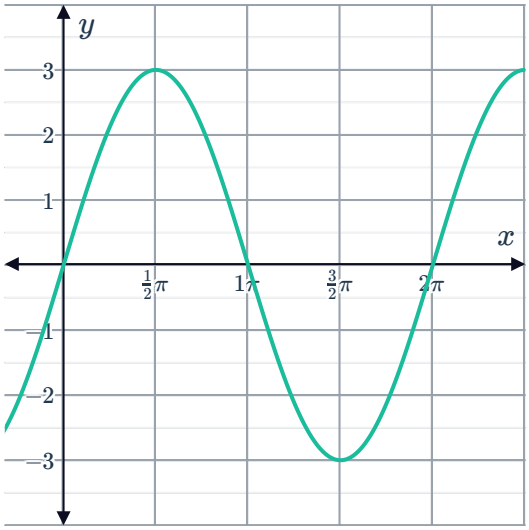
c



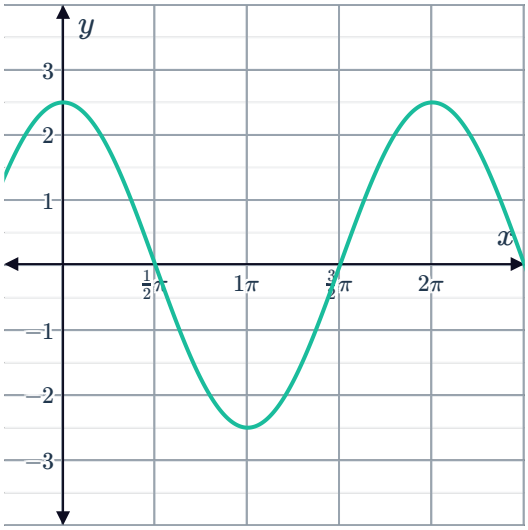
d



e

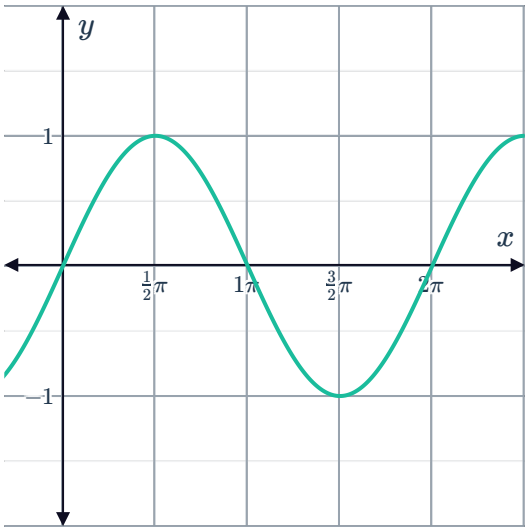


f



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6 Consider the graph of the function $y = \sin x$ for $0 \leq x < 2\pi$:
At which value of x in the given domain would $y = -\sin x$ have a maximum value?



7 Consider the function $f(x) = -8\sin x$, where $0 \leq x \leq \pi$.

a State the amplitude of the function.

b Find the value of $f(\pi)$.

c Find the minimum value of the function.

8 The function $y = k\sin x$ has a maximum value of 5. Find the value of k, where $k > 0$.

9 A sine function has the form $y = c\sin x$, a range of $[-2, 2]$ and a minimum at $\frac{\pi}{2}$. Find an expression for y.

10 For each of following functions:

i State the amplitude.

ii Sketch a graph of the function.

a $y = 3\sin x$.

b $y = 4\cos x$

c $y = \frac{4}{3}\cos x$

d $y = -5\cos x$

e $y = \frac{5}{4}\sin x$

f $y = -4\sin x$

11 Consider the function $y = -3\cos x$.

a State the maximum value of the function.

b State the minimum value of the function.

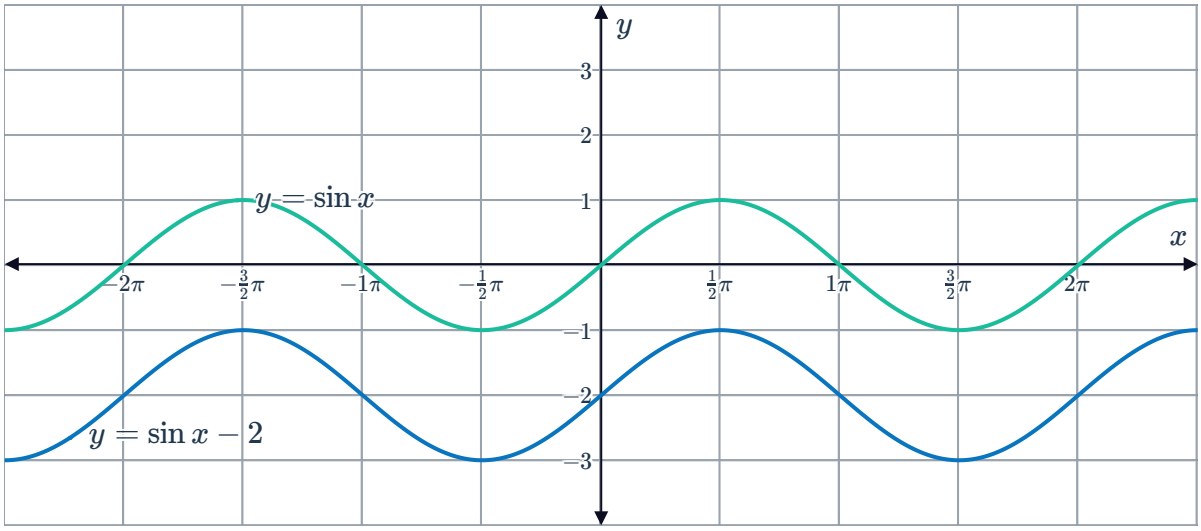
c State the amplitude of the function.

d Describe the two transformations required to obtain the graph of $y = -3\cos x$ from the graph of $y = \cos x$.

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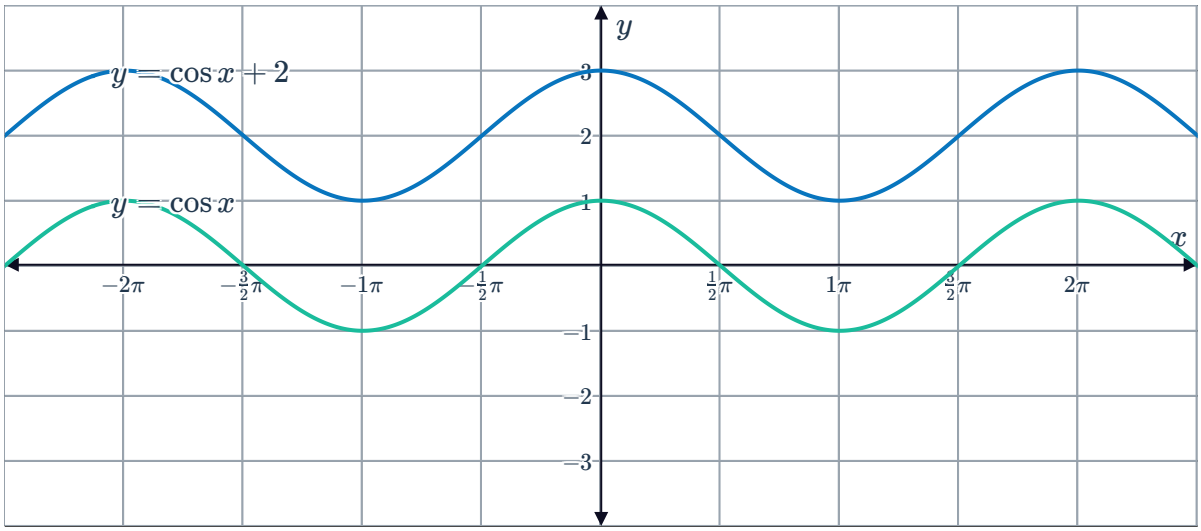
Vertical translations of sine and cosine functions

12 Consider the two graphs $y = \sin x$ and $y = \sin x - 2$ below:



Describe the transformation required to obtain the graph of $y = \sin x - 2$ from $y = \sin x$.

13 Consider the two graphs $y = \cos x$ and $y = \cos x + 2$ below:



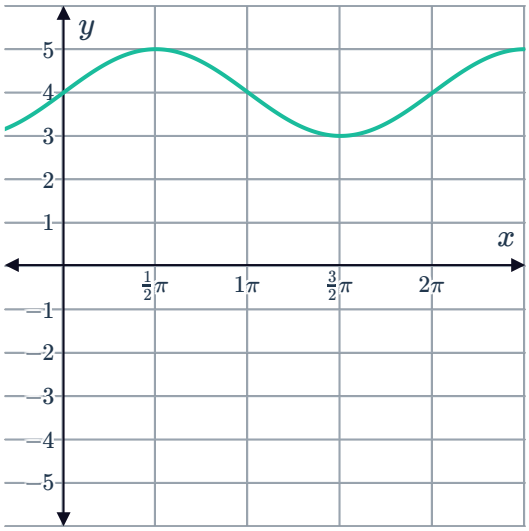
Describe the transformation required to obtain the graph of $y = \cos x + 2$ from $y = \cos x$.

14 Describe the transformation required to obtain the graph of $y = \sin x + 4$ from $y = \sin x$.

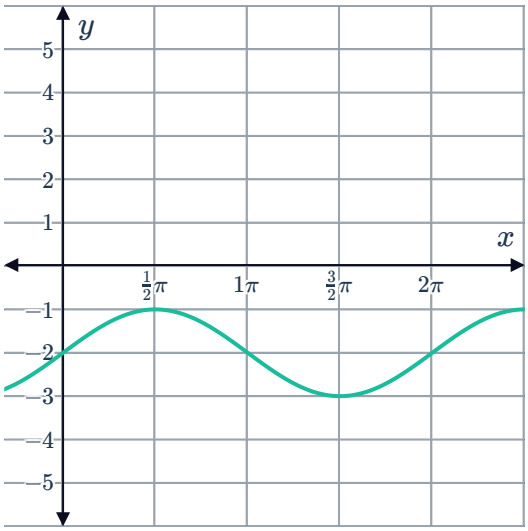
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15 State the equation of each of the functions graphed below given that they are of the form $y = \sin x + k$:

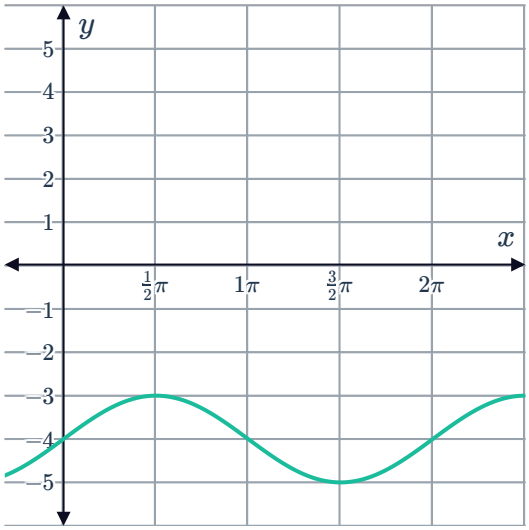
a



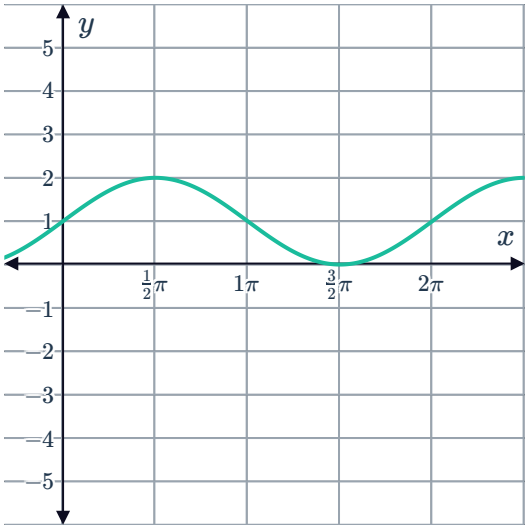
b



c



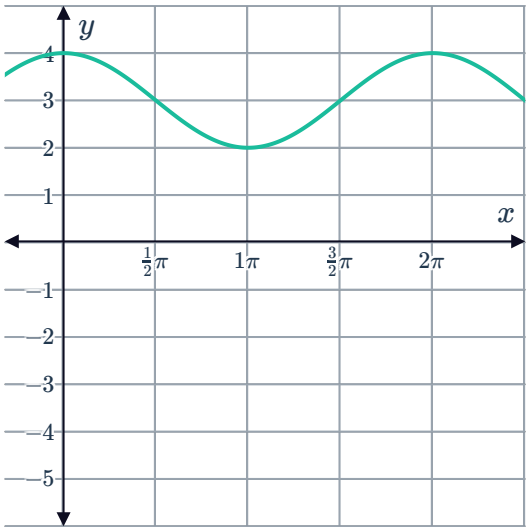
d



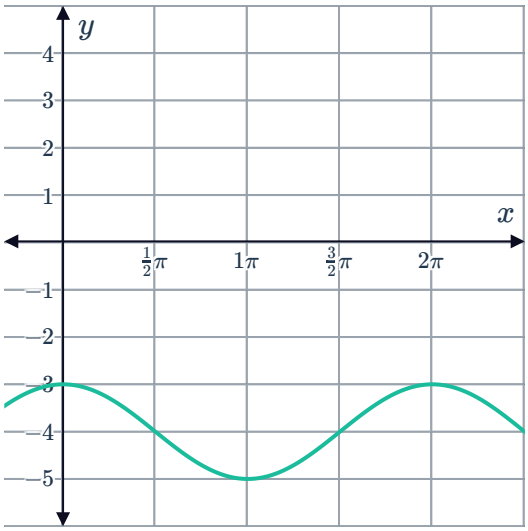
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16 State the equation of each of the functions graphed below given that they are of the form $y = \cos x + k$:

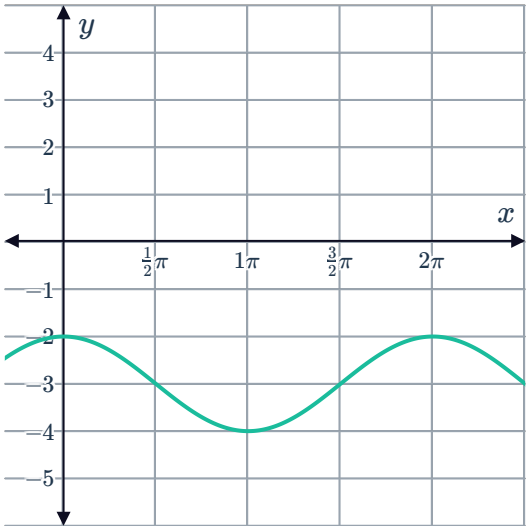
a



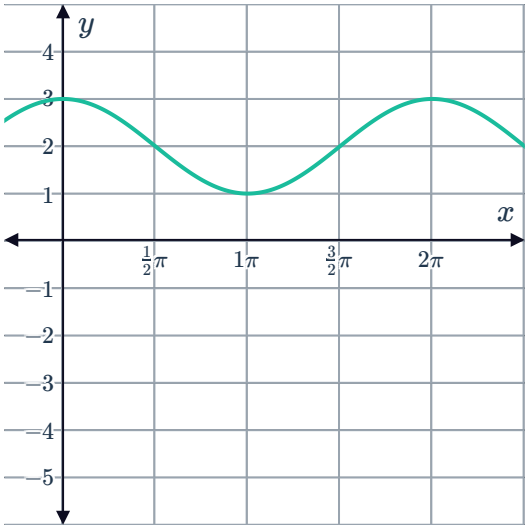
b



c



d



17 Describe the two transformations required to turn the graph of $y = \cos x$ into the graph of $y = -\cos x + 3$.

18 State the equation of the line that the function $y = \sin x + 2$ oscillates about.

19 Consider the function $y = \cos x + 4$.

a Describe the transformation required to obtain the graph of $y = \cos x + 4$ from $y = \cos x$.

b State the period of the function, giving your answer in degrees.

c Find the maximum value of the function.

d Find the minimum value of the function.

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20 For each of the functions below:

i State the period of the function, giving your answer in radians.

ii State the amplitude of the function.

iii Find the maximum value of the function.

iv Find the minimum value of the function.

v Sketch a graph of the function for $-2\pi \leq x \leq 2\pi$.

a $y = \sin x + 1$ **b** $y = \cos x - 3$ **c** $y = \sin x - 2$ **d** $y = 3\sin x + 2$

e $y = 3\cos x - 3$ **f** $y = 2\sin x - 3$

21 For each of the functions below:

i Find the value of y when $x = \frac{\pi}{2}$.

ii State the amplitude of of the function.

iii Find the maximum value of the function.

iv Find the minimum value of the function.

v Describe the transformations required to obtain the graph of the function from $y = \cos x$.

vi Sketch a graph of the function for $0 \leq x \leq 2\pi$.

a $y = 2\cos x + 3$ **b** $y = 2\cos x - 3$ **c** $y = -4\cos x + 2$ **d** $y = \frac{1}{3}\sin x + 2$

22 The function $y = \cos x + 5$ is translated 4 units up.

a State the equation of the new function after the translation.

b Find the maximum value of the new function.

23 For each of the functions below:

i Find the maximum value of the function. ii Find the minimum value of the function.

a $y = 2 - 3\sin x$ **b** $y = 4 - 3\sin x$ **c** $y = (\sin x)^2 + 1$ **d** $y = \frac{4}{3\cos x + 5}$

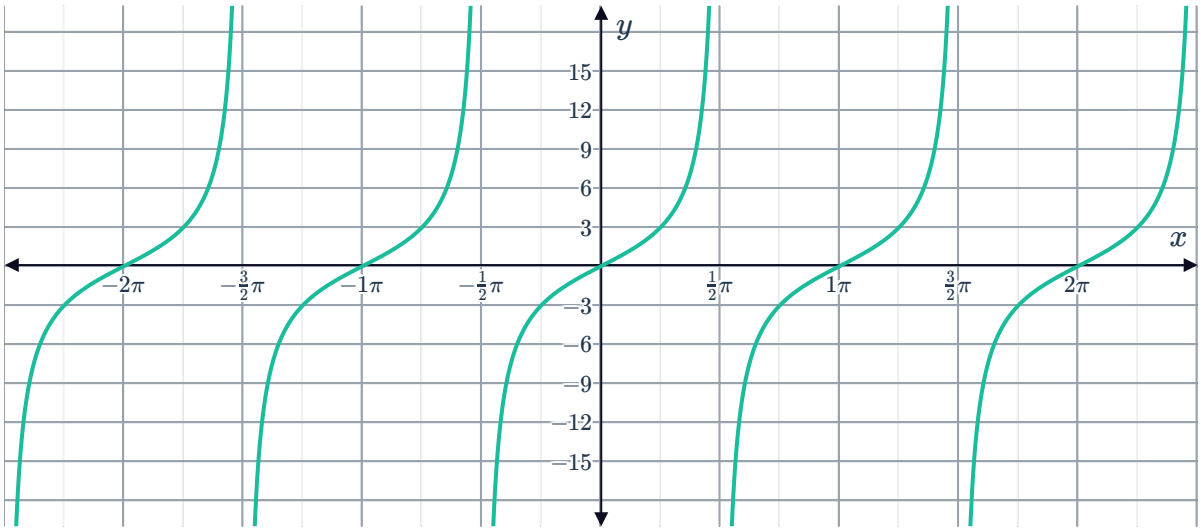
24 A sine function, y, has the form $y = c\sin x + d$ and a range of $[0, 4]$. Find an expression for y, where $c > 0$.

25 A cosine function, y, has the form $y = c\cos x - d$ and a range of $[-10, 6]$. Find an expression for y, where $c > 0$.

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The tangent function and vertical dilations

26 Consider the given graph of $f(x) = 3\tan x$:

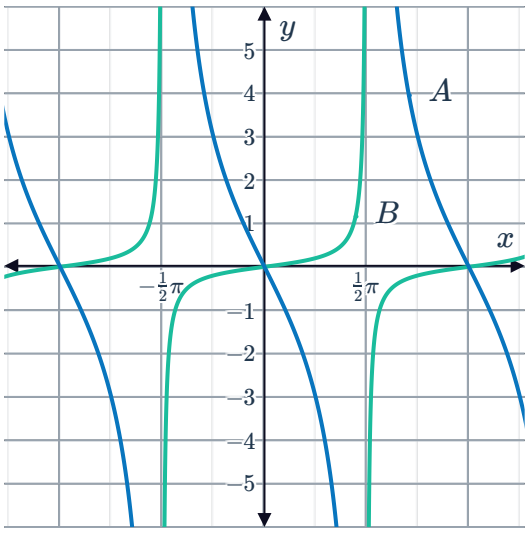


a State the domain of $f(x)$.

b State the range of $f(x)$.

c The graph of $f(x)$ has its domain restricted to $[0, \pi)$, state the range of the restricted graph.

27 Consider the functions $f(x) = \frac{1}{5}\tan x$ and $g(x) = -3\tan x$ drawn on the given set of axes:
Which function has been labelled:



a A **b** B

28 For each of the functions below:

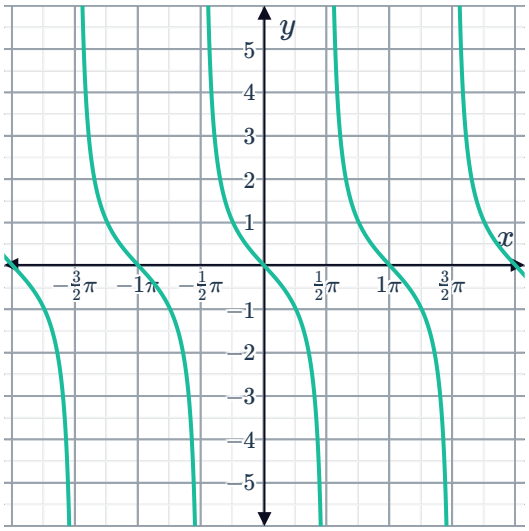
a State the vertical dilation factor required to obtain the given function from $y = \tan x$.

b State whether the graph would appear stretched or compressed in relation to $y = \tan x$.

a $y = \frac{1}{4}\tan x$ **b** $y = 6\tan x$ **c** $y = \frac{5}{3}\tan x$ **d** $y = 0.8\tan x$

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29 Consider the given graph of a function of the form $y = a\tan x$:

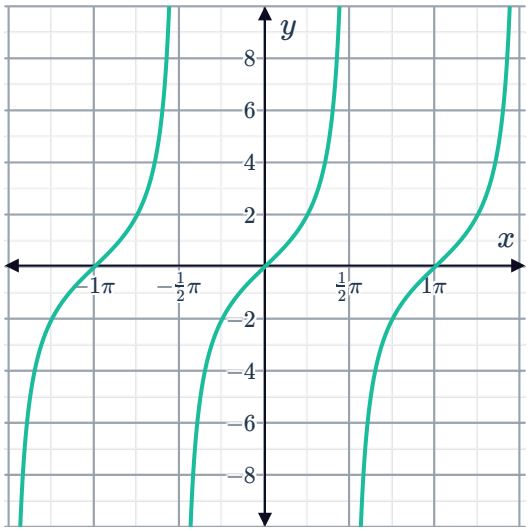


a Describe the transformation required to obtain the given function from $y = \tan x$.

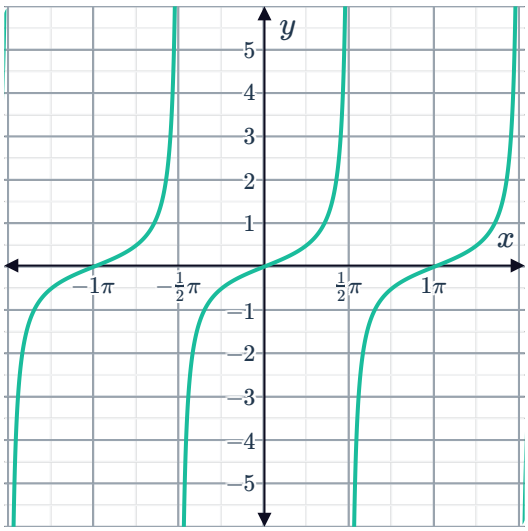
b State the equation of the graphed function.

30 State the equation of each of the functions graphed below given that they are of the form $y = a\tan x$:

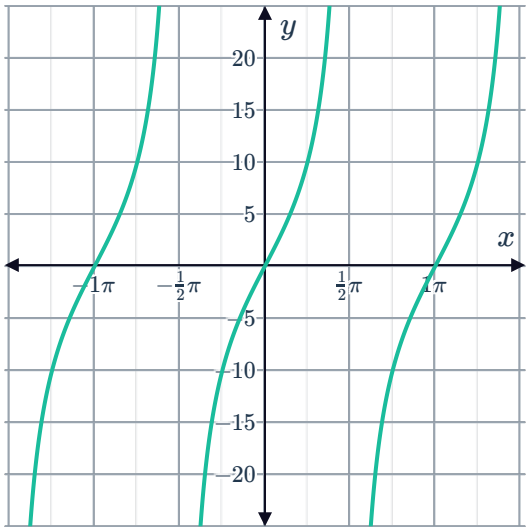
a



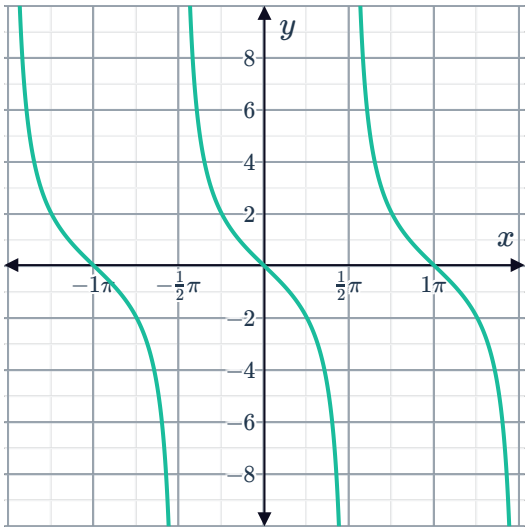
b



c

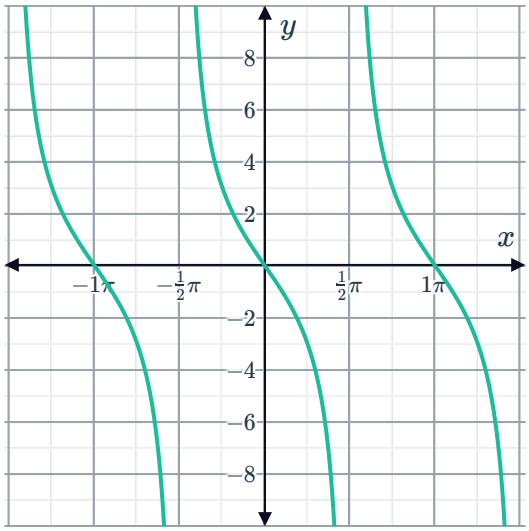


d

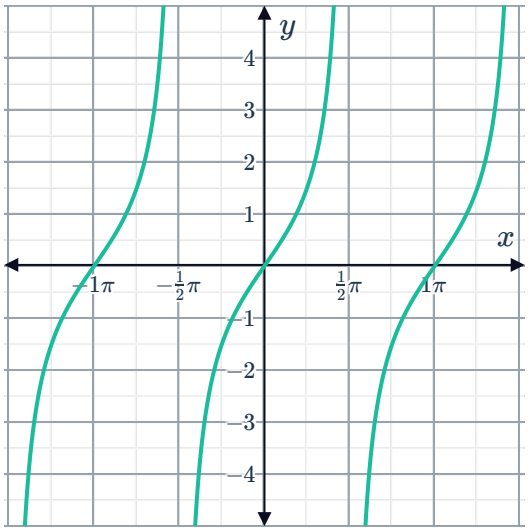


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e



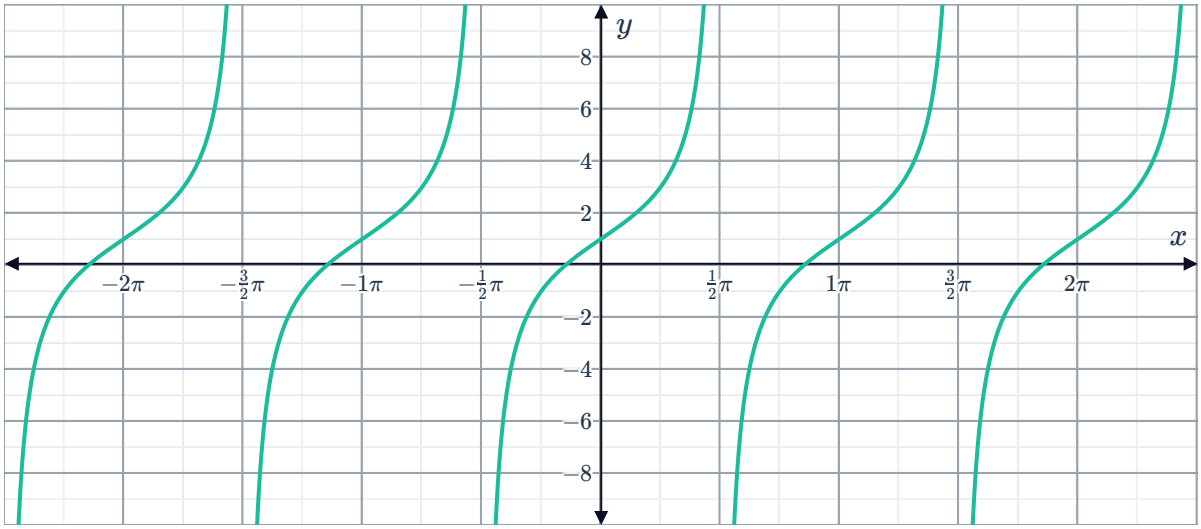
f



31 On the same set of axes, sketch the graph of $y = \tan x$ and $y = \frac{1}{2}\tan x$.

Vertical translations of the tangent function

32 Consider the given graph of $f(x) = 2\tan x + 1$:

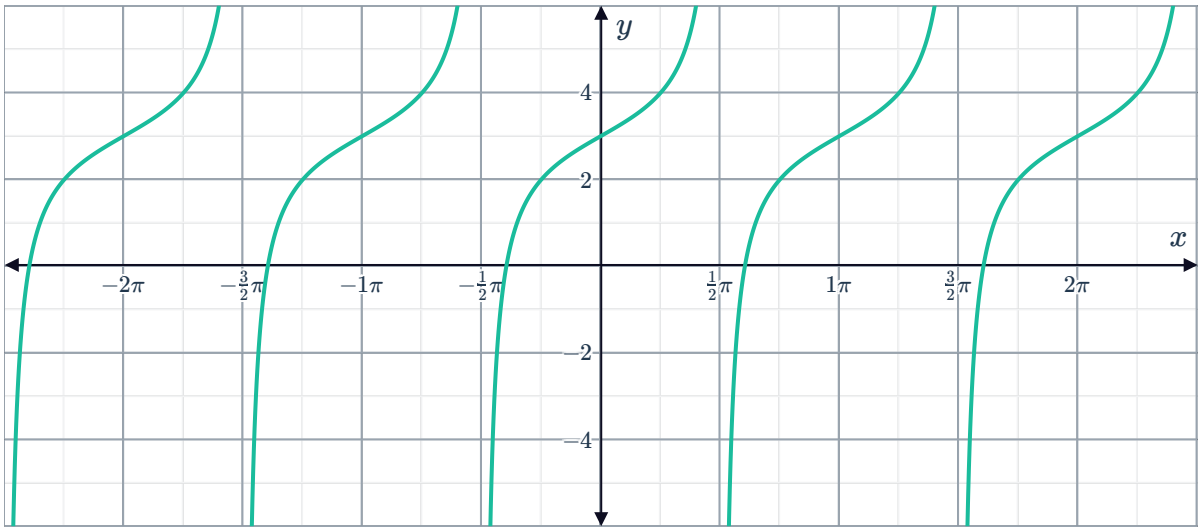


a State the domain of $f(x)$.

b State the range of $f(x)$.

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33 Consider the given graph of $f(x) = \tan x + 3$:



a State the domain of $f(x)$.

b State the range of $f(x)$.

c The graph of $f(x)$ has its domain restricted to $\left[0, \frac{\pi}{2}\right)$, state the range of the restricted graph.

34 For each of the functions below:

i Find the y-intercept.

ii Find the value of y when $x = \frac{\pi}{4}$.

iii Find the period of the function.

iv Find the distance between the asymptotes of the function.

v State the equation of the first asymptote of the function for $x \geq 0$.

vi State the equation of the first asymptote of the function for $x \leq 0$.

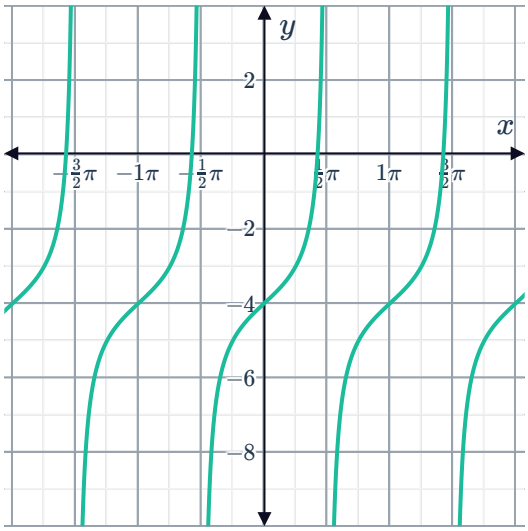
vii Sketch a graph of the function.

a $y = \tan x - 2$

b $y = 5\tan x + 3$

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35 Consider the given graph of a tangent function:



a State the transformation required to obtain the given function from $y = \tan x$

b State the equation of the graphed function.

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